FastSpeech 2 Text-to-Speech Synthesis

English and Telugu Technical Report

Implementation • Dataset preparation • Training • Inference • Variance-adaptor analysis

Abstract

This work implements two separate FastSpeech 2 text-to-speech systems, one for English using Svarah and one for Telugu using IndicTTS Telugu. The implementation follows the non-autoregressive FastSpeech 2 pipeline: text/phoneme encoding, duration–pitch–energy prediction through the variance adaptor, length regulation, mel-spectrogram decoding, PostNet refinement, and waveform synthesis using a universal HiFi-GAN vocoder. Montreal Forced Aligner (MFA) is used to obtain forced-alignment information for duration supervision. The report documents the data preparation, training configuration, generated test-sample statistics, variance-adaptor behaviour, cross-language differences, limitations, and reproducibility steps.

The English report records 30,000 training steps. The Telugu run was resumed from step 16,000 after correcting validation pitch and energy normalization.

1. Implementation approach

1.1 FastSpeech 2 architecture

The systems are based on the reference implementation by ming024/FastSpeech2. FastSpeech 2 is a non-autoregressive acoustic model that predicts a mel-spectrogram from phoneme-level text representations. Its principal components are:

- Text/phoneme encoder: converts the input phoneme sequence into hidden representations using FFT (Feed-Forward Transformer) blocks.
- Variance adaptor: contains predictors for phoneme duration, fundamental frequency (F0/pitch), and energy. These prosodic features are used to enrich the encoder representations.
- Length regulator: expands each phoneme representation according to its predicted duration, producing a frame-level sequence.
- Mel-spectrogram decoder: maps the expanded sequence to a mel-spectrogram.
- PostNet: refines the predicted mel-spectrogram.
- Vocoder: a universal HiFi-GAN generator converts the predicted mel-spectrogram into a time-domain waveform.

MFA forced alignment supplies timing/alignment information used to derive duration targets. Pitch and energy are extracted at phoneme level and normalized using statistics computed from the training split. The same broad acoustic architecture is used for both languages, while text representation, phoneme conversion, normalization and dataset handling are language-specific.

1.2 Language-specific adaptation

Area	English system	Telugu system
Dataset	Svarah	IndicTTS Telugu
Text/phoneme representation	English phoneme inventory with ARPAbet-style symbols and grapheme-to-phoneme (G2P) handling, as recorded in the report.	Telugu-specific phone mapping/normalization; the report describes a rule-based phone conversion path.
Alignment	MFA forced alignment; one utterance was dropped after alignment.	MFA forced alignment; 42 utterances were dropped in the initial report statistics. The later selected feature pack contains 2,616 train and 653 validation utterances.
Key challenges	Accent, speaker and recording-channel variation; lexical stress and pronunciation of long/OOV words.	Telugu orthography-to-phone mapping, conjuncts/long words, homographs, word boundaries and number/date normalization.

2. Dataset selection and preprocessing

Telugu

Train: 2063 utts / 2.96 h; validation: 479 utts / 0.69 h; 2 speaker(s); 42 utterances dropped after alignment. Utterance-disjoint split, seed 1234.

1. Training configuration

Batch 16, Adam (0.9, 0.98), warm-up 4000, 30000 steps, LR annealing at 65% and 85% of training, grad-clip 1.0, hidden 256, 4 encoder / 6 decoder FFT layers.

2. Loss curves

Telugu: 30000 steps, best validation total loss 3.7463 at step 1000 (last val 3.9657, last train 0.7655).

English

Train: 3594 utts / 4.60 h; validation: 771 utts / 1.02 h; 114 speaker(s); 1 utterances dropped after alignment. Utterance-disjoint split, seed 1234.

1. Training configuration

Batch 16, Adam (0.9, 0.98), warm-up 4000, 30000 steps, LR annealing at 65% and 85% of training, grad-clip 1.0, hidden 256, 4 encoder / 6 decoder FFT layers.

2. Loss curves

English: 30000 steps, best validation total loss 3.4678 at step 2000 (last val 3.7065, last train 0.9998)

2.1 Dataset selection and split

Language	Training split	Validation split	Speakers	Split / alignment notes
English (Svarah)	3,594 utterances; 4.60 h	771 utterances; 1.02 h	114	Utterance-disjoint; seed 1234; 1 utterance dropped after alignment.
Telugu (IndicTTS Telugu) — initial report	2,063 utterances; 2.96 h	479 utterances; 0.69 h	2	Utterance-disjoint; seed 1234; 42 utterances dropped after alignment.
Telugu — final selected feature pack	2,616 utterances; 5.300 h selected audio	653 utterances; 1.301 h selected audio	2	Later audited selection; no skipped feature items; train/validation disjoint.

The final Telugu selected feature pack is the authoritative dataset-duration result for the corrected run. It supersedes the smaller initial Telugu split counts in the early report. The selected duration exceeds the approximate 5 h/1 h targets while maintaining an utterance-disjoint split. English provides approximately 4.60 h/1.02 h, which is close to the requested 5 h/1 h but is slightly below

five training hours.

2.2 Selection and preprocessing pipeline

- Subsampling: utterances were selected using the language-specific data settings in each config.yaml. The split uses seed 1234 and separates utterances between training and validation.
- Filtering: the report records duration filtering in the 1.5–12 second range and word-count filters. Retain the exact thresholds from the committed config.yaml as the reproducibility source of truth.
- Audio preparation: audio is resampled to 22.05 kHz and peak-normalized before feature extraction.
- Text preparation: text is normalized and converted to the language-specific phone sequence. Preserve the supplied test text; where the assignment text itself contains unusual spacing or spelling, record the exact form used for inference.
- Alignment: MFA is used to obtain phone/word timing and duration supervision. Utterances without usable alignment are excluded.
- Acoustic features: mel-spectrogram, phoneme-level F0/pitch and energy features are extracted. Pitch and energy are normalized using training-set mean and standard deviation; validation features must use the same training statistics.
- Quality checks: the final Telugu feature audit confirmed 2,616 training and 653 validation utterances, 5.300 h and 1.301 h respectively, with no skipped feature items and no train/validation overlap.

Telugu validation normalization correction: during the resumed Telugu experiment, the validation pitch and energy arrays were initially left in raw scale while the model targets were normalized. This produced extremely large validation pitch/energy losses (tens of thousands) and made those early validation results invalid. The validation arrays were then normalized once using the training statistics from stats.json. After correction, validation pitch had mean 0.0755 and standard deviation 0.9291; validation energy had mean 0.0519 and standard deviation 1.0319. Subsequent validation metrics are the ones reported in Section 4. The pre-correction Telugu validation curve must not be used as evidence of model performance.

3. Training configuration

Parameter	Configuration recorded in reports
Optimizer	Adam, $\beta1 = 0.9$, $\beta2 = 0.98$
Batch size	16
Warm-up	4,000 steps
Planned training length	30,000 steps
Learning-rate schedule	Annealing at 65% and 85% of training
Gradient clipping	1.0
Hidden dimension	256
FFT blocks	4 encoder layers; 6 decoder layers
Training targets	Mel-spectrogram, duration, phoneme-level pitch/F0 and energy

4. Training and validation loss analysis

4.1 English

The English reports 30,000 training steps, a best validation total loss of 3.4678 at step 2,000, a final validation total loss of 3.7065, and a final training total loss of 0.9998. The training curve drops sharply during the early optimization phase and then decreases more gradually, indicating rapid initial fitting followed by slower refinement. The validation curve is comparatively flat and fluctuates around the mid-3 range after its early minimum. The persistent train–validation gap suggests limited generalization and possible overfitting or a train/validation distribution mismatch. The validation curve does not show sustained improvement in later training, so the best-validation checkpoint is more defensible than selecting solely by the final training loss.

4.2 Telugu

The early Telugu records a 30,000-step run with best validation loss 3.7463 at step 1,000, final validation loss 3.9657 and final training loss 0.7655. However, those validation results were generated before the validation pitch and energy normalization issue was corrected. They are not valid for the corrected experiment and are intentionally not used to assess convergence.

In the corrected run, training loss decreased from approximately 0.84 around step 16,000 to 0.7579 at step 22,000. At step 22,000, validation total loss was 1.7861, with mel 0.5755, postnet 0.5753, pitch 0.3960, energy 0.1864 and duration 0.0529. The corrected validation losses are finite and no longer dominated by raw-scale pitch/energy targets. Validation remains higher than training, particularly for pitch and energy, indicating a generalization gap. Since the latest supplied post-fix result is at step 22,000, a final 30,000-step convergence claim cannot yet be made.

4.3 Interpretation and caveats

- Both systems show substantial early reduction in training loss followed by slower improvement.
- Validation losses are higher than training losses in both reports, consistent with a generalization gap. This may reflect limited data, speaker/channel variation, alignment noise, or overfitting; loss curves alone cannot isolate the cause.
- The English validation curve is relatively flat after its early best value. The corrected Telugu validation value is usable but still higher than training loss.

5.1 English test sentences

ID / text	Duration /	Pitch	Energy	Plot-based interpretation

	phones			
E1 — The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	5.41 s; 85 phones; 0 predicted pauses	F0 136–170 Hz; mean 148 Hz	Energy 5.06–30.67	The duration plot distributes time across a long 85-phone sequence; SH, NG and Z are among the longest phones (104 ms). F0 varies within a relatively narrow range, with local rises and falls rather than a flat contour. Energy varies by phone, indicating non-uniform predicted prominence. The long, morphologically complex OOV word is a stress/G2P challenge; inspect its WAV for syllable deletion or stress errors.
E2a — The orchestra's conduct during the ceremony was praised by the audience.	3.48 s; 50 phones; 0 predicted pauses	F0 131–182 Hz; mean 146 Hz	Energy 4.33–31.69	The duration plot shows variable phone durations, with EY1 and S among the longest (128 ms) and Z at 116 ms. The pitch curve contains a local rise around the early-middle portion and falls thereafter. Energy has sharp phone-level variation. This sentence tests the noun reading of "conduct"; verify that the generated pronunciation reflects the intended noun sense.
E2b — The researchers will conduct an experiment to evaluate the proposed method.	3.82 s; 56 phones; 0 predicted pauses	F0 131–159 Hz; mean 142 Hz	Energy 4.03–34.17	ER0 is among the longest phones (128 ms), followed by CH and TH (116 ms). The pitch contour is comparatively restrained, with modest local movement. Energy peaks are distributed across the sentence. This sentence contrasts with E2a by using "conduct" as a verb; a text-only phoneme pipeline may not resolve the homograph reliably without context-aware G2P.
E3 — The rhyth m of the orchestra changed unexpectedly.	2.57 s; 37 phones; 0 predicted pauses	F0 134–156 Hz; mean 142 Hz	Energy 6.26–39.70	AY1 is the longest phone (139 ms), followed by EY1 (128 ms) and IY0 (116 ms). The pitch contour contains several local excursions, then remains lower toward the end. Energy varies strongly at the beginning and is more even later. The supplied test string contains the unusual spacing "rhyth m"; preserve and report the exact input used, and note that this may affect tokenization/G2P.
E4 — The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	5.55 s; 78 phones; 1 predicted pause	F0 135–162 Hz; mean 147 Hz	Energy 4.17–30.73	AY1 is among the longest phones (139 ms); F and AY1 are also prolonged (116 ms). The contour is relatively narrow, with local changes around phrase segments. Energy is variable, including stronger early peaks and a more moderate later region. This sentence tests time, date and the token "andaroo"; number verbalization and unfamiliar-token pronunciation should be checked directly in the audio.

5.2 Telugu test sentences

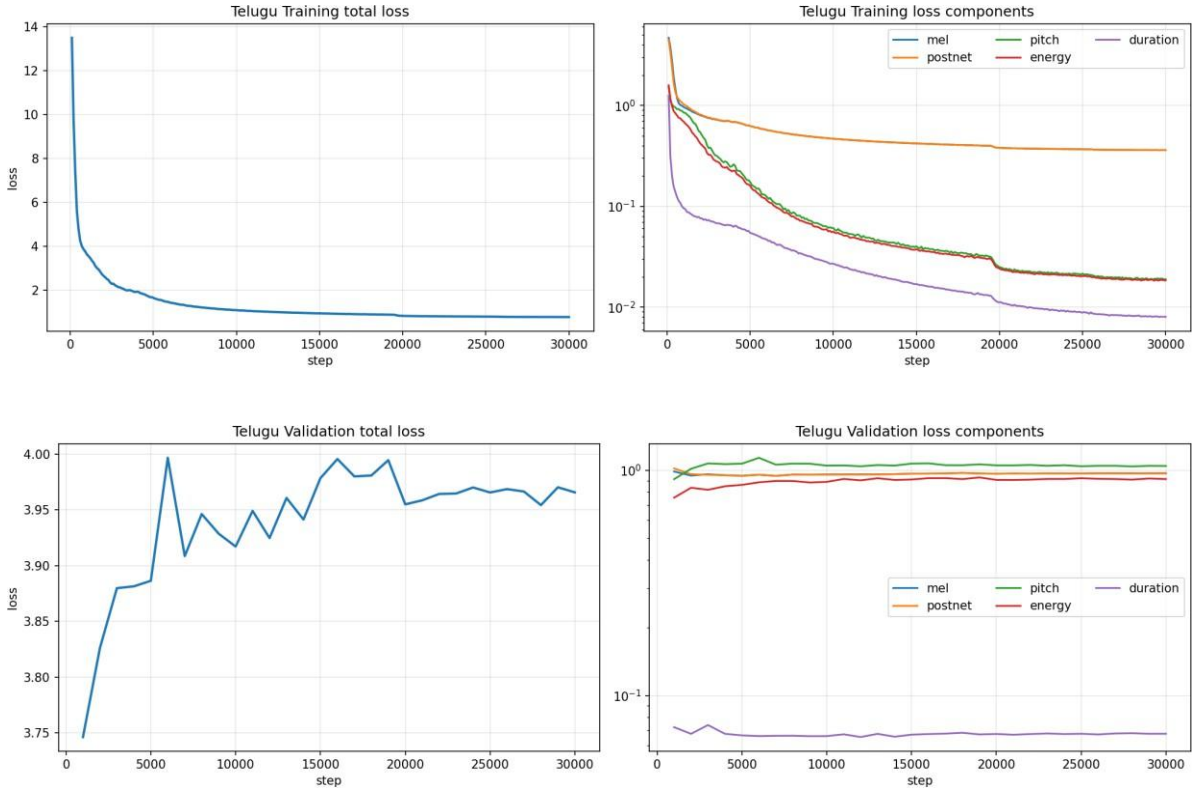
ID / text	Duration / phones	Pitch	Energy	Plot-based interpretation
T1 — న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.	4.93 s; 66 phones + 5 pauses	F0 147–282 Hz; mean 204 Hz	Energy 13.57–117.94	The sequence is relatively long and contains five predicted pauses. The longest predicted units include ఓ (128 ms) and శ (116 ms). F0 spans a broad range, with higher local peaks in the early portion and a more moderate contour later. Energy has strong early peaks and becomes more even across later phones. The long technical loanword is a likely pronunciation/G2P stress test; inspect syllable realization in the WAV.
T2a — తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	2.54 s; 31 phones + 5 pauses	F0 150–240 Hz; mean 195 Hz	Energy 4.94–113.61	The short sentence has five predicted pauses, a notable proportion of the total phone sequence. Repeated long vowel units ఆ appear among the longest predictions (128/116 ms). The pitch contour rises early and then trends downward. Energy shows high early peaks and lower values through parts of the later sequence. This sentence uses పట్టు in the grip/hold sense.
T2b — ఆమె పట్టుచీరను ధరించి వేడుకకు వచ్చింది.	2.81 s; 37 phones + 4 pauses	F0 164–261 Hz; mean 203 Hz	Energy 12.33–110.35	The duration profile includes a prominent long ఆ (151 ms), followed by చ (116 ms) and a pause (104 ms). Pitch varies over a wider range than the English samples, with an early rise followed by a gradual decline. Energy is strongest in selected early phones and then becomes more moderate. This is the contrasting “silk saree” use of పట్టు; it is a useful test of lexical and phonetic handling of the same written form in different contexts.
T3 — పులి అడవిలో నిశ్శబ్దంగా నడిచింది.	2.16 s; 31 phones + 3 pauses	F0 166–284 Hz; mean 202 Hz	Energy 13.66–117.50	The sentence has three predicted pauses and a short total duration. శ is the longest listed phone (139 ms), followed by చ (116 ms) and ఆ (104 ms). The F0 plot shows a pronounced early rise and peak, followed by a lower, steadier contour. Energy contains strong isolated peaks and lower-energy intervals. The quiet-walking semantic content does not guarantee low acoustic energy; the model predicts energy from learned acoustic patterns, not sentence meaning.
T4 — ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.	8.42 s; 119 phones + 17 pauses	F0 188–246 Hz; mean 202 Hz	Energy 14.67–93.79	This is the longest test sample and has 17 predicted pauses, consistent with a long sequence containing a date, time and multiple phrase boundaries. ఈ is among the longest units (151 ms), with ఐ also listed at 139 and 104 ms. Pitch remains within a narrower range than T1–T3 and varies more gradually. Energy has strong early peaks and lower, more stable levels in later regions. Verify date/time verbalization and pause placement by listening.

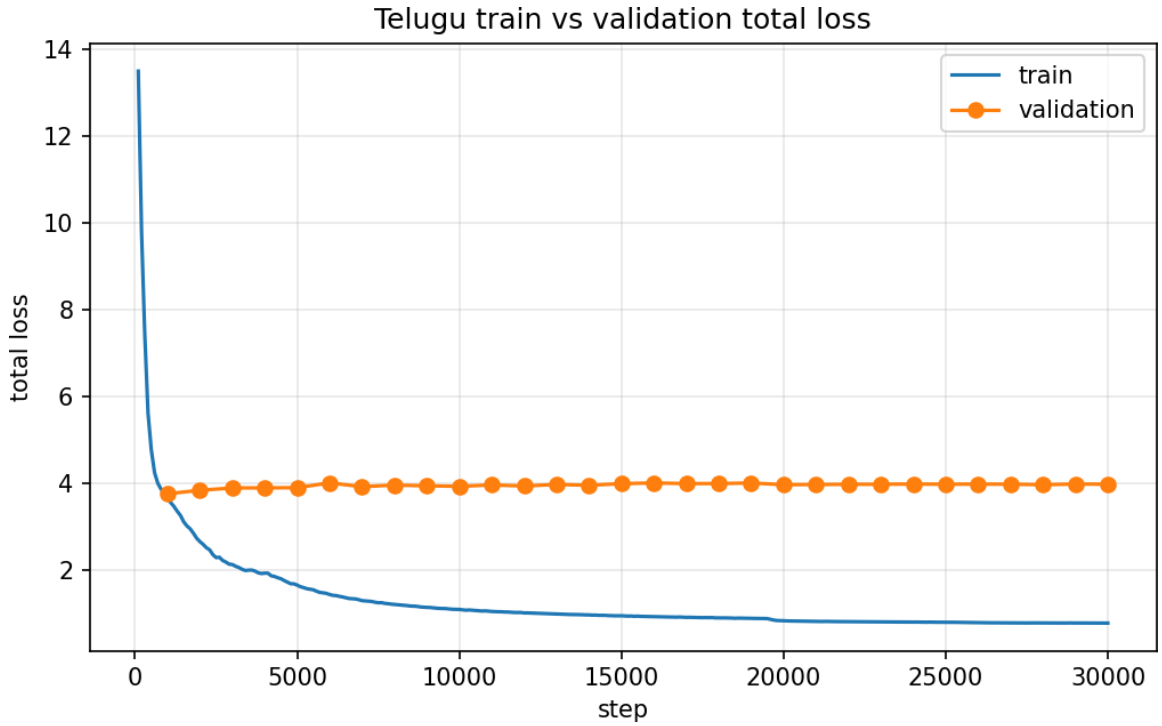
6. English–Telugu comparison, limitations and improvements

Dimension	English / Svarah	Telugu / IndicTTS Telugu
Data characteristics	Larger speaker diversity in the supplied split (114 speakers); the report notes accent, channel and partly spontaneous variation.	Two speakers in the supplied statistics; comparatively controlled studio-style recordings are noted in the report.
Text front end	English phoneme/ARPAbet-style representation and G2P; stress and homograph resolution are challenges.	Telugu-specific phone mapping and normalization; conjuncts, long words, sandhi/word boundaries and script-to-phone coverage are challenges.
Observed test challenges	Long OOV word (antidisestablishmentarianism), noun/verb homograph "conduct", unusual "rhyth m" spacing, and date/time/"andaroo" handling.	Technical loanword in T1, homograph పట్టు in T2a/T2b, and date/time verbalization in T4.
Loss interpretation	Training decreases but validation remains relatively flat after its early best point; generalization gap is present.	Corrected validation is usable after normalization fix; at step 22k it remains above training loss, especially for pitch and energy.
Acoustic outputs	Phone-level F0 mean is approximately 142–148 Hz across these samples; durations range 2.57–5.55 s.	Phone-level F0 mean is approximately 195–204 Hz; durations range 2.16–8.42 s. These ranges are model outputs and should not be interpreted as a direct voice-quality comparison.

6.1 Limitations

- The English and Telugu datasets differ in speaker count, recording conditions and language structure, so direct loss comparisons are not controlled.
- G2P and text normalization can fail on out-of-vocabulary words, homographs, spelling anomalies, numerals, dates and times.
- MFA alignment errors can create noisy duration targets, especially for uncommon words or mismatched phone inventories.
- The universal HiFi-GAN vocoder may not be optimally matched to every speaker, dataset or predicted mel distribution.
- Validation performance is weaker than training performance, suggesting a generalization gap.


Telugu Training total loss
loss
step
Telugu Training loss components
mel
pitch
duration
postnet
energy
step
Telugu Validation total loss
loss
step
Telugu Validation loss components
mel
pitch
duration
postnet
energy
step

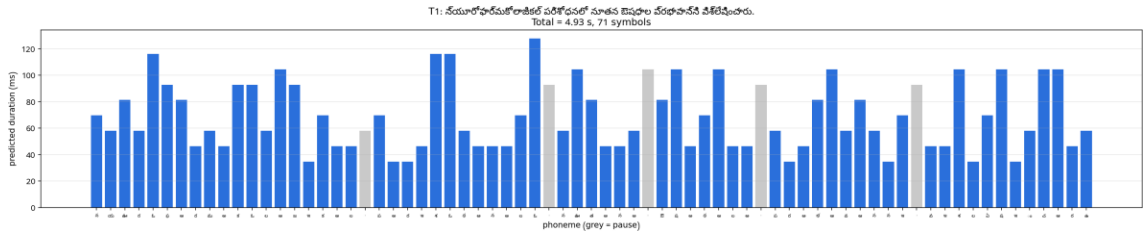

Telugu train vs validation total loss
train
validation
total loss
step

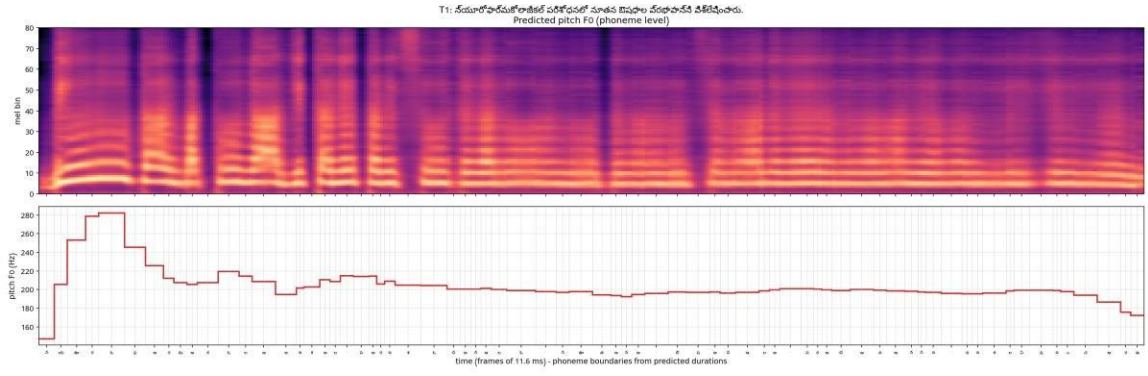
3. Generated speech and variance-adaptor analysis (per test sentence)

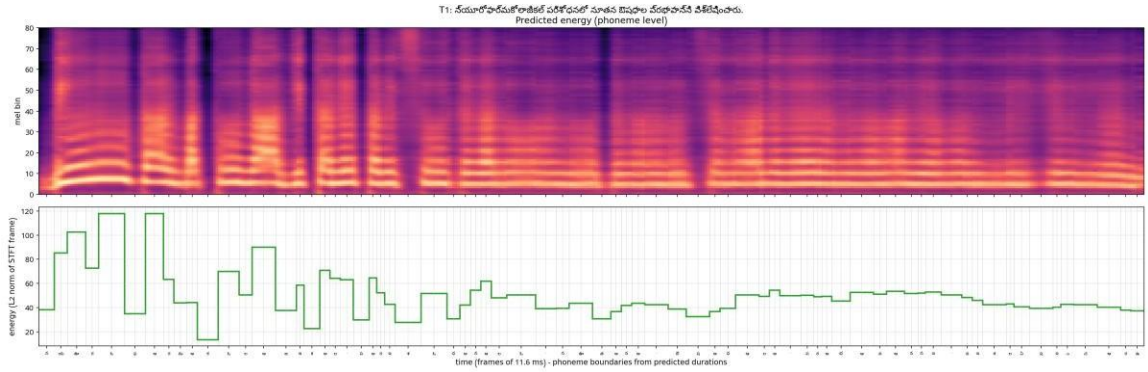
Telugu

T1: న్యూ ఢిల్లీ కోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

Duration 4.93 s, 66 phones + 5 pauses; F0 147-282 Hz (mean 204); energy 13.57-117.94; longest: ఓ (128 ms), శ (116 ms), ఓ (116 ms)

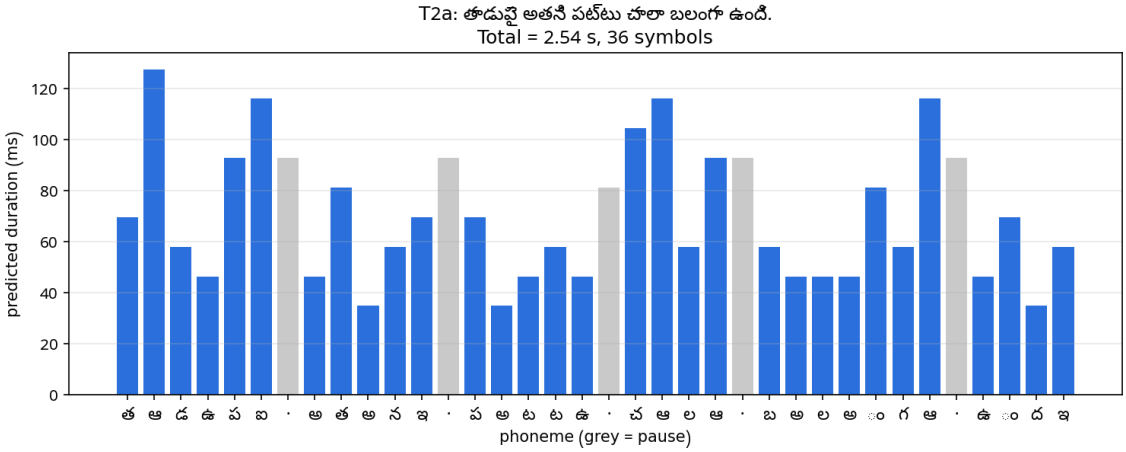


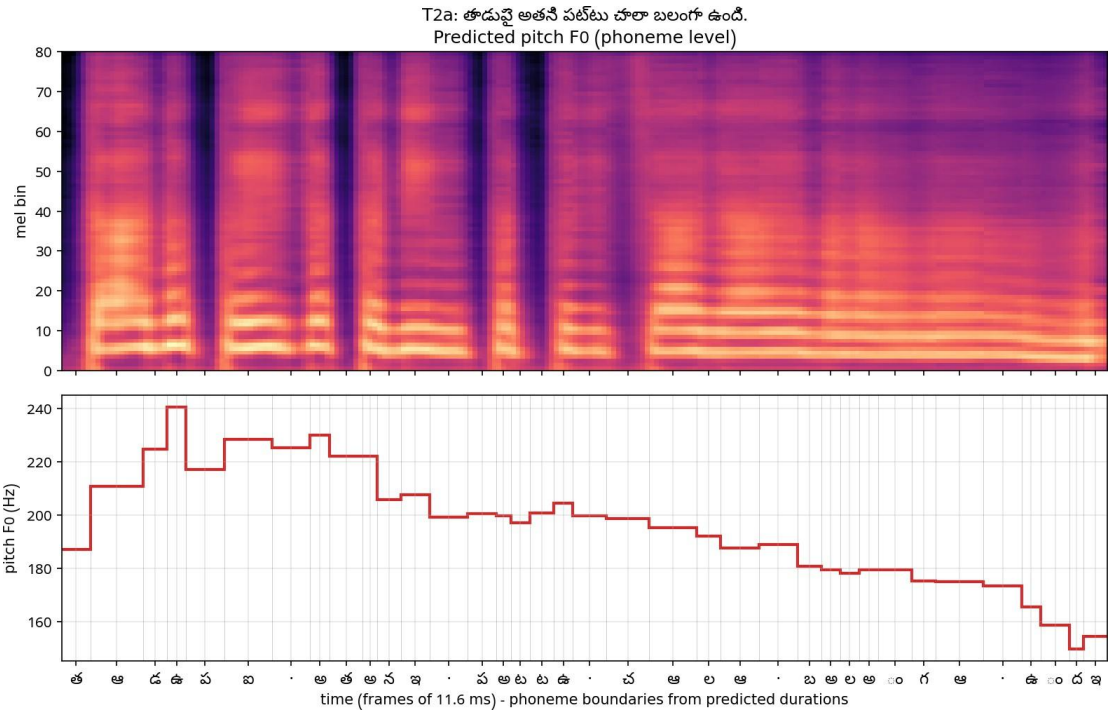


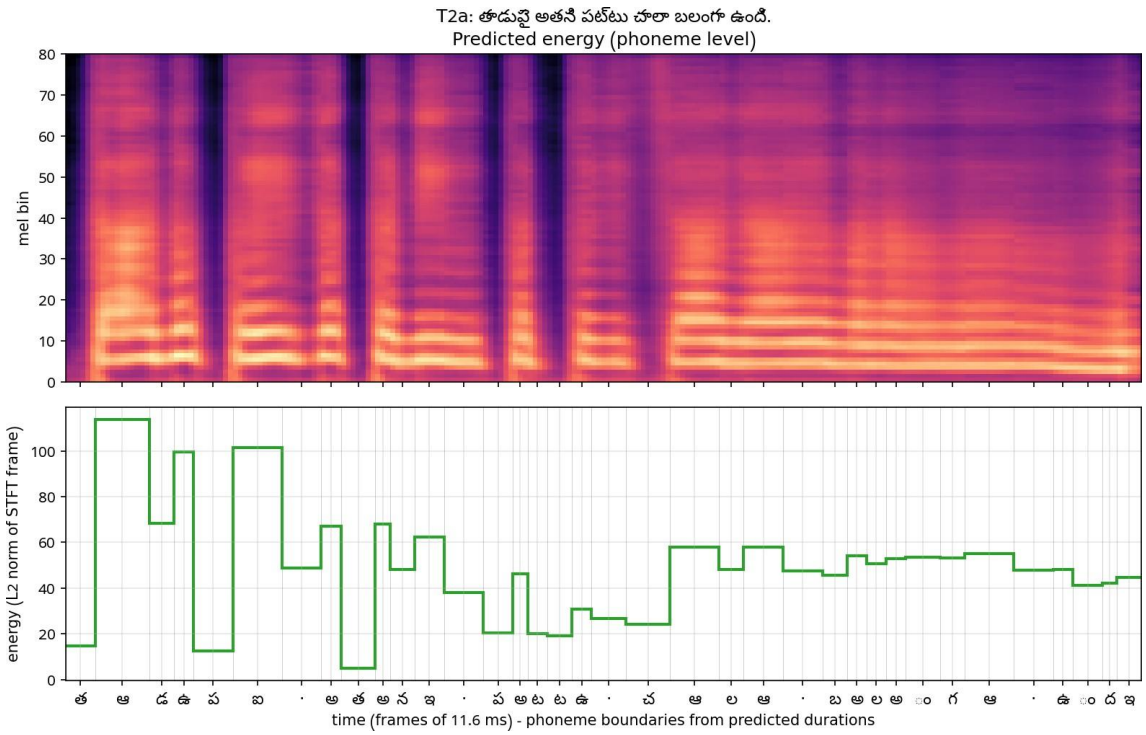


T2a: తాడుపై అతని పట్టు చాలా బలంగా ఉంది.

Duration 2.54 s, 31 phones + 5 pauses; F0 150-240 Hz (mean 195); energy 4.94-113.61; longest: ఆ (128 ms), ఆ (116 ms), ఆ (116 ms)

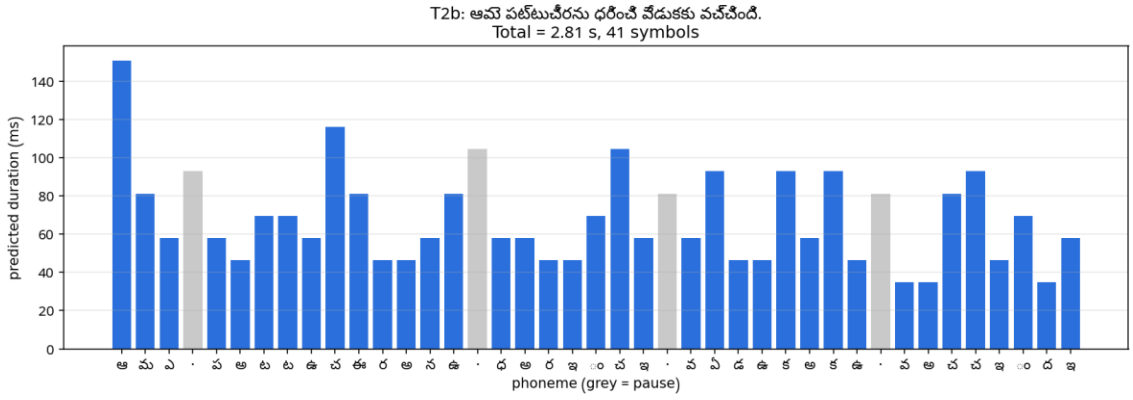

T2a: తూడుపై అతని పట్టు చూలా బలంగా ఉంది.
Total = 2.54 s, 36 symbols
predicted duration (ms)
0
20
40
60
80
100
120
phoneme (grey = pause)

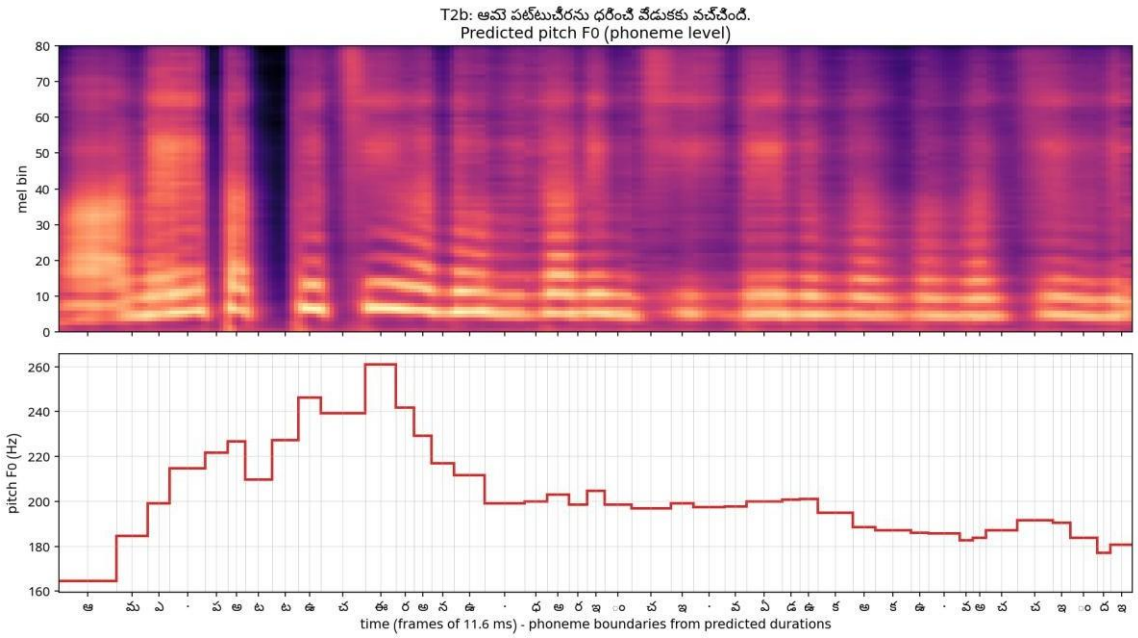

T2a: తూడుపై అతని పట్టు చూలా బలంగా ఉంది.
Predicted pitch F0 (phoneme level)
mel bin
0
10
20
30
40
50
60
70
80
pitch F0 (Hz)
160
180
200
220
240
time (frames of 11.6 ms) - phoneme boundaries from predicted durations

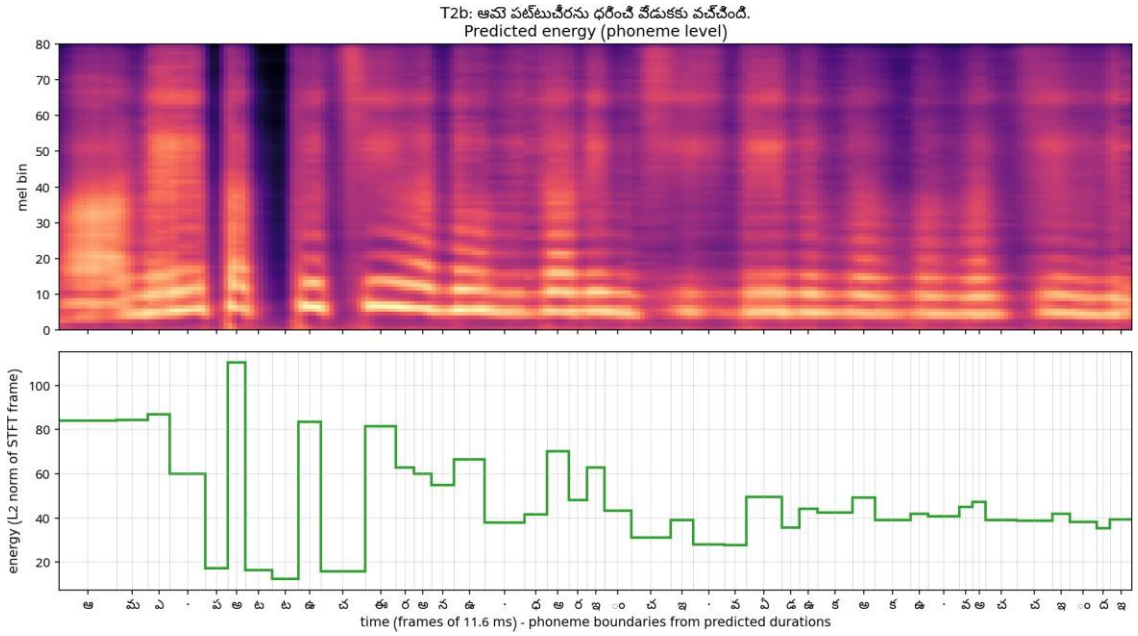


T2b: ఆమె పట్టుచీరను ధరించి వేడుకకు వచ్చ ింది.

Duration 2.81 s, 37 phones + 4 pauses; F0 164-261 Hz (mean 203); energy 12.33-110.35; longest: ఆ (151 ms), చ (116 ms), sp (104 ms)

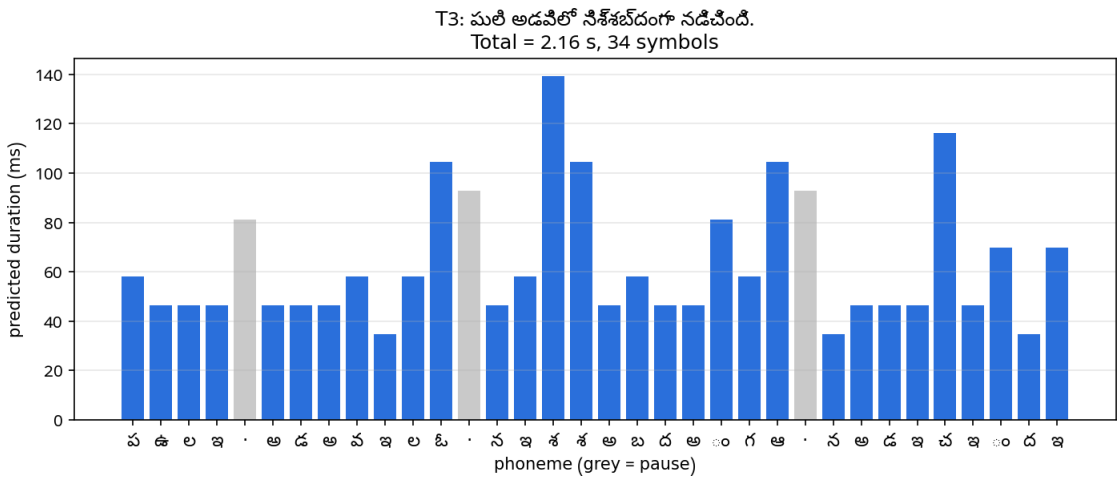


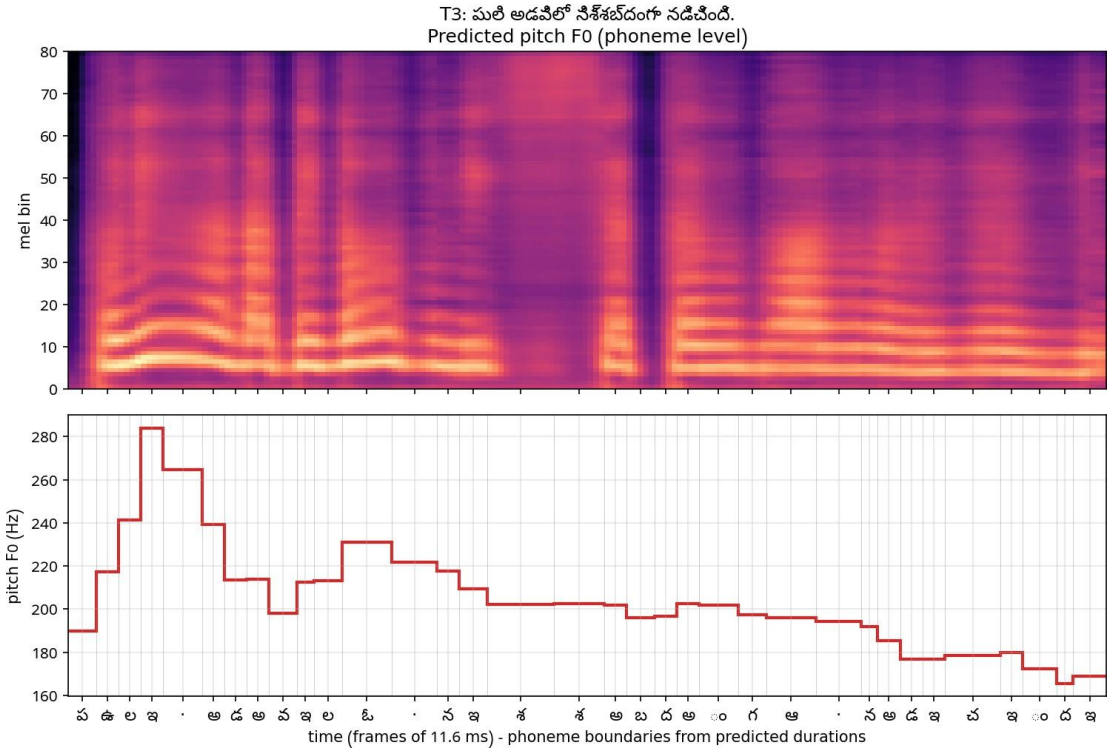


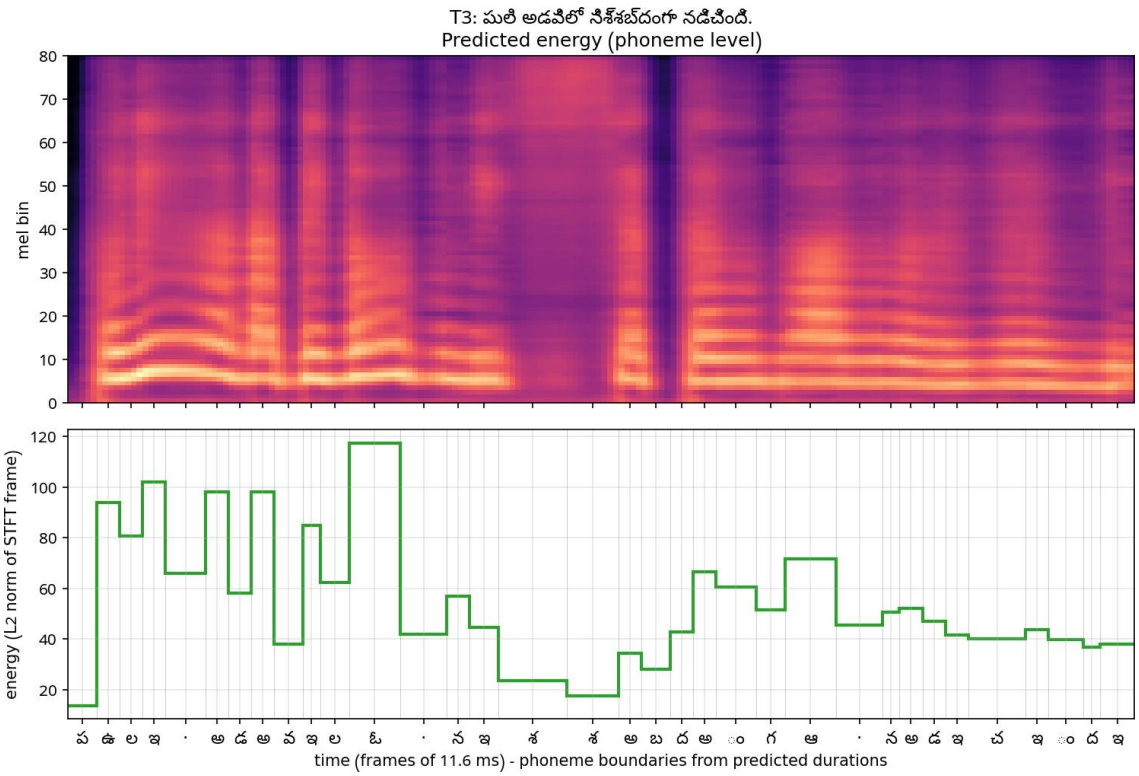


T3: పులి అడవిలో నిశ్శ బ్దంగానడిచింది.

Duration 2.16 s, 31 phones + 3 pauses; F0 166-284 Hz (mean 202); energy 13.66-117.50; longest: శ (139 ms), చ (116 ms), ఆ (104 ms)

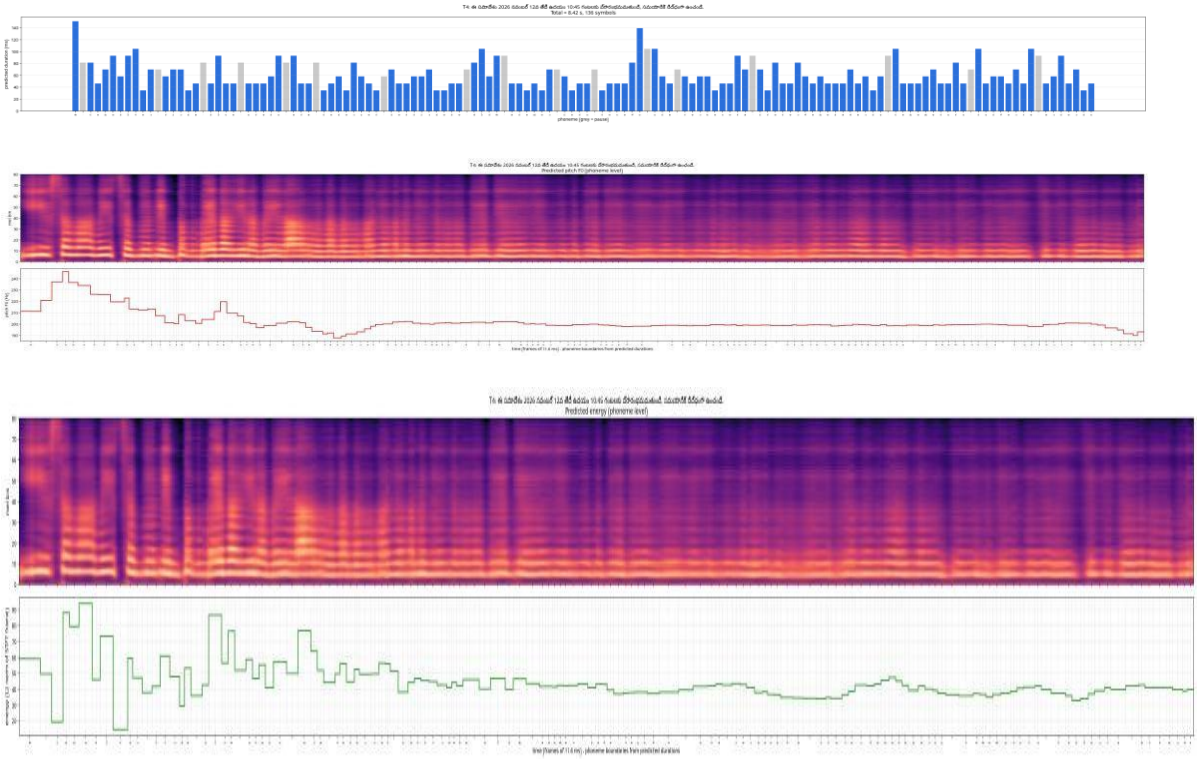

T3: పులి అడవిలో నిశ్శబ్దంగా నడిచింది.
Total = 2.16 s, 34 symbols
predicted duration (ms)
phoneme (grey = pause)


T3: పులి అడవిలో నిశ్శబ్దంగా నడిచింది.
Predicted pitch F0 (phoneme level)
mel bin
pitch F0 (Hz)
time (frames of 11.6 ms) - phoneme boundaries from predicted durations

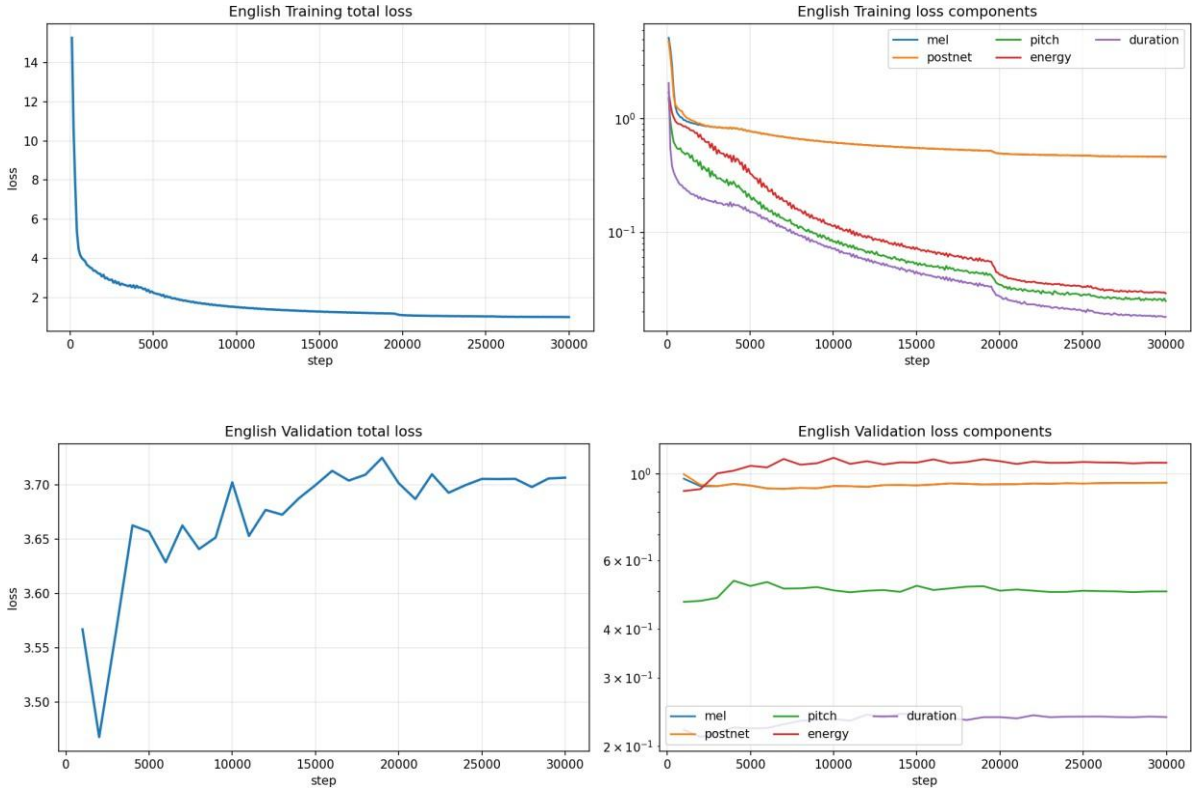


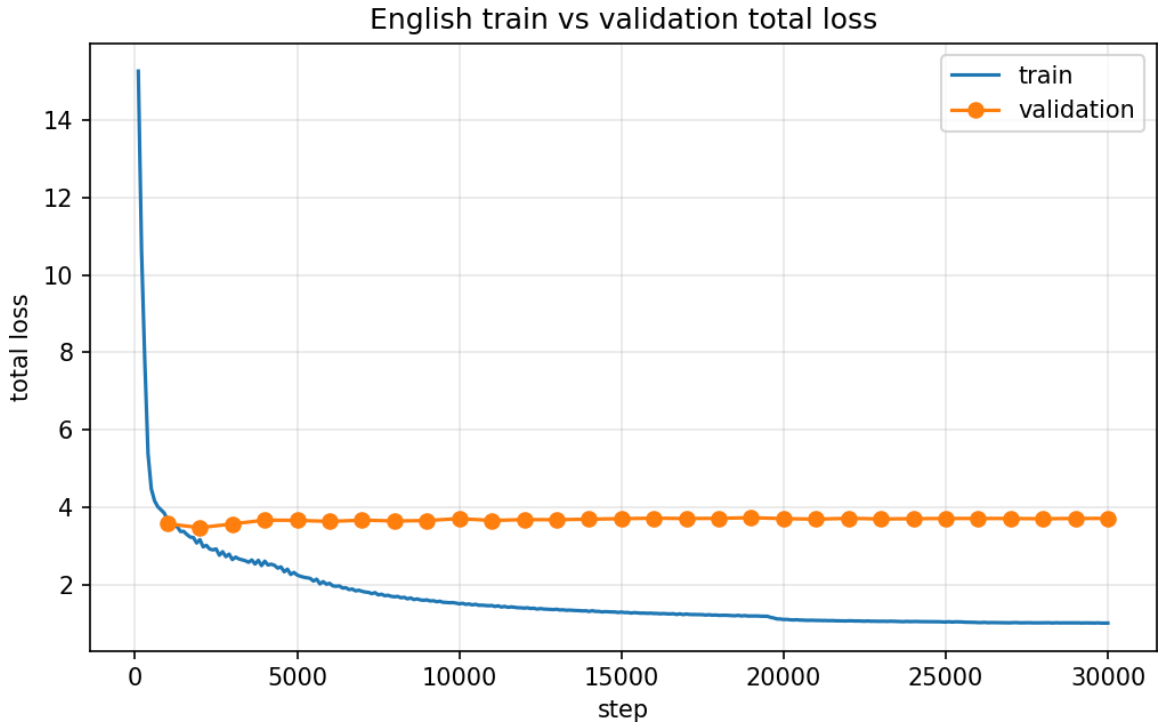
T4: ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

Duration 8.42 s, 119 phones + 17 pauses; F0 188-246 Hz (mean 202); energy 14.67-93.79; longest: ఈ (151 ms), ఐ (139 ms), ఐ (104 ms)



E


English Training total loss
loss
step
English Training loss components
mel
postnet
pitch
energy
duration
step
English Validation total loss
loss
step
English Validation loss components
mel
postnet
pitch
energy
duration
step

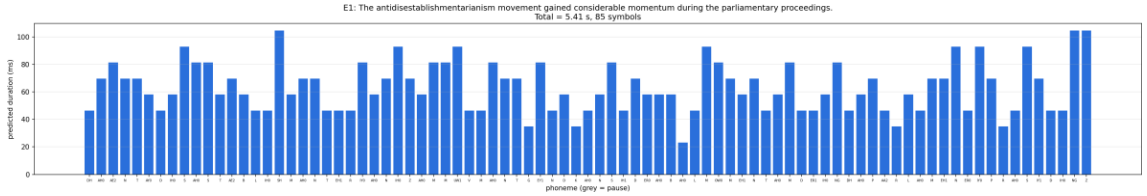

English train vs validation total loss
train
validation
total loss
step

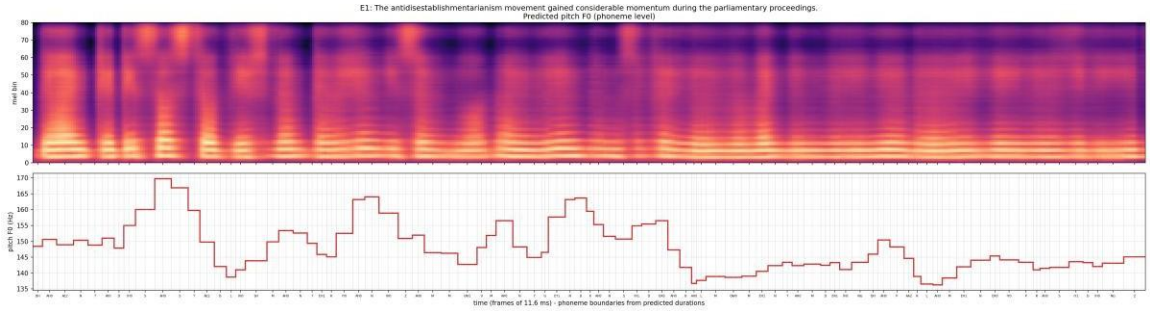
3. Generated speech and variance-adaptor analysis (per test sentence)

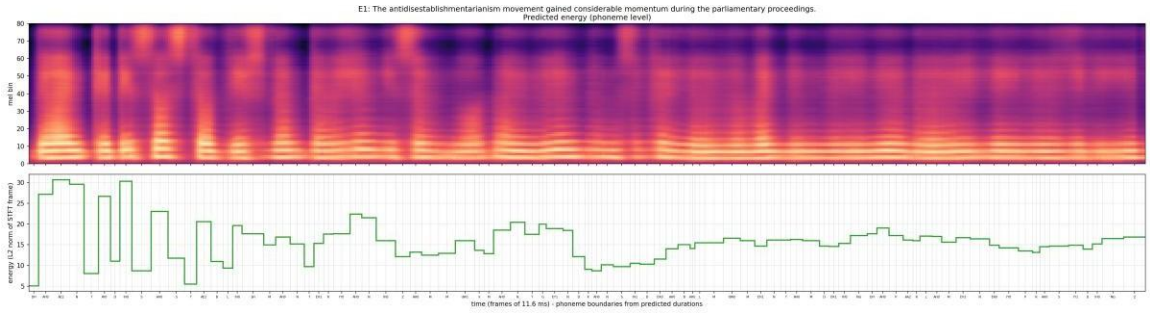
English

E1: The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.

Duration 5.41 s, 85 phones + 0 pauses; F0 136-170 Hz (mean 148); energy 5.06-30.67; longest: SH (104 ms), NG (104 ms), Z (104 ms)

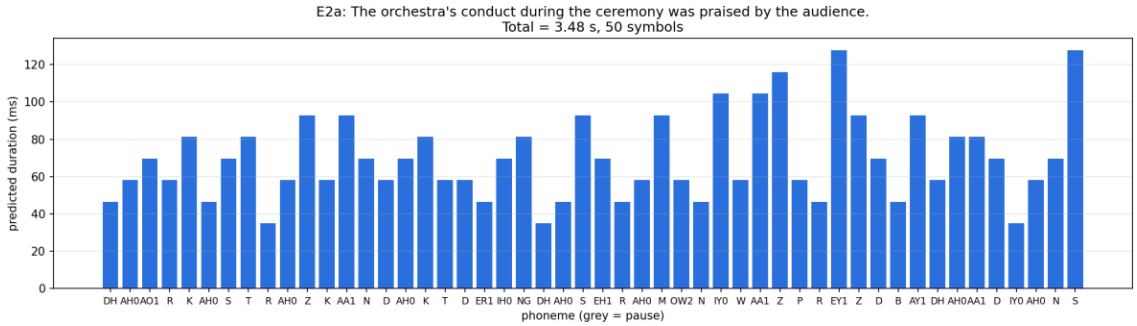


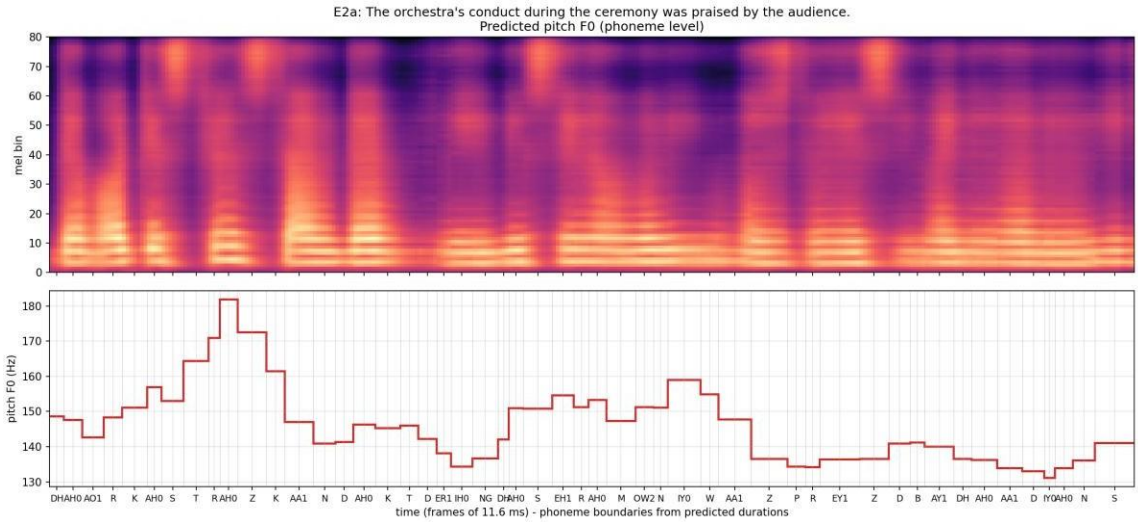


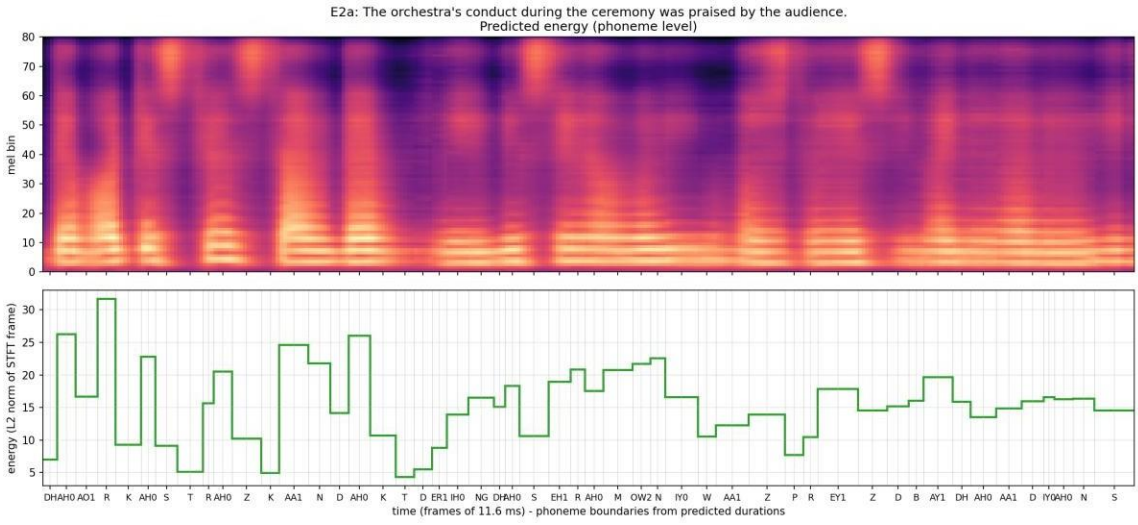


E2a: The orchestra's conduct during the ceremony was praised by the audience.

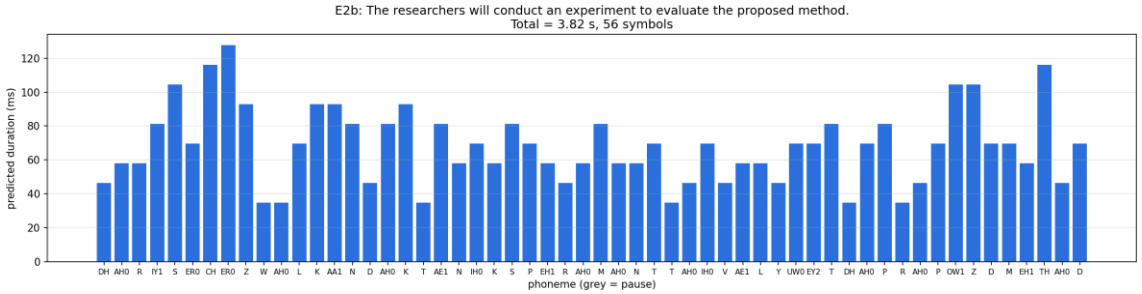
Duration 3.48 s, 50 phones + 0 pauses; F0 131-182 Hz (mean 146); energy 4.33-31.69; longest: EY1 (128 ms), S (128 ms), Z (116 ms)

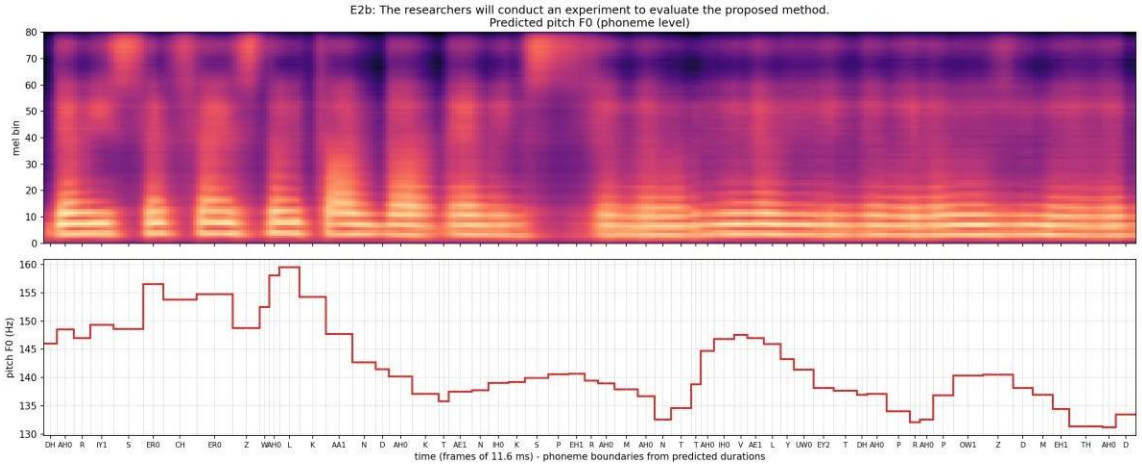


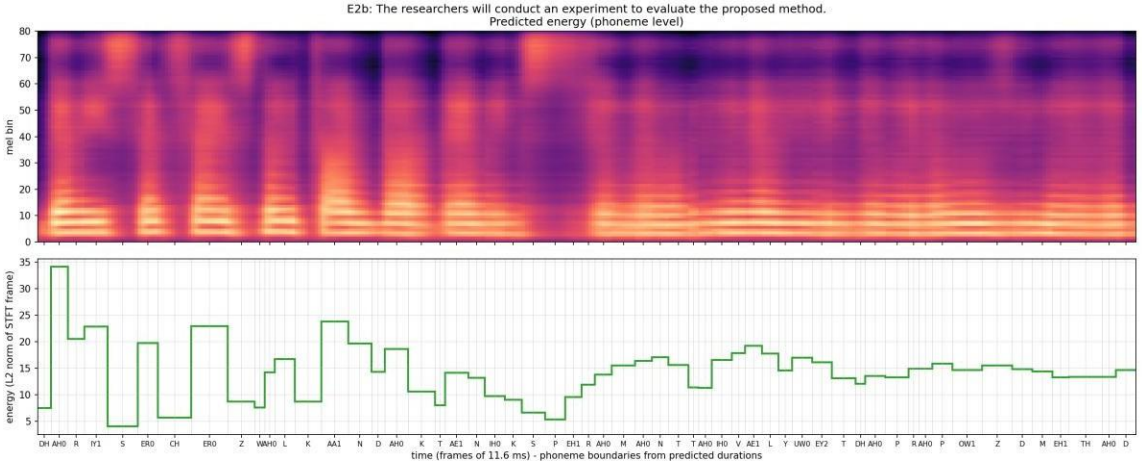




E2b: The researchers will conduct an experiment to evaluate the proposed method. Duration 3.82 s, 56 phones + 0 pauses; F0 131-159 Hz (mean 142); energy 4.03-34.17; longest: ER0 (128 ms), CH (116 ms), TH (116 ms)

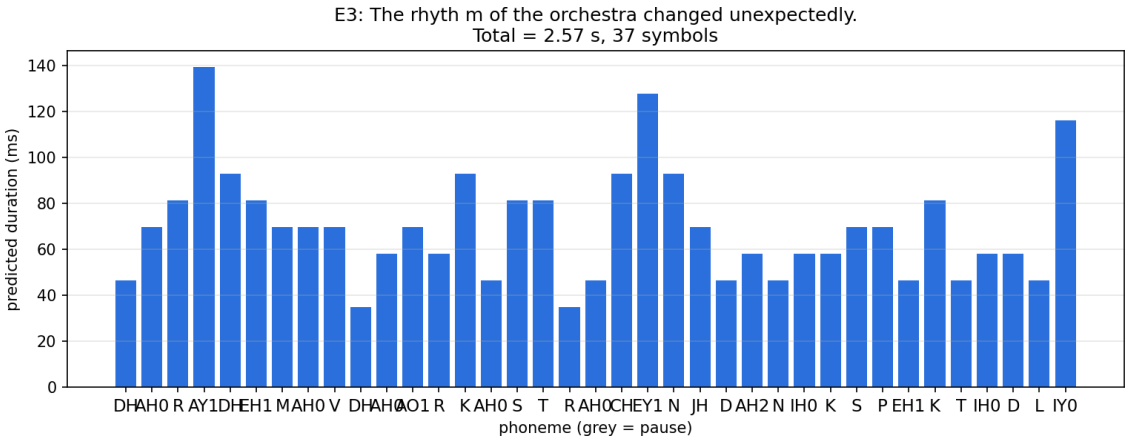


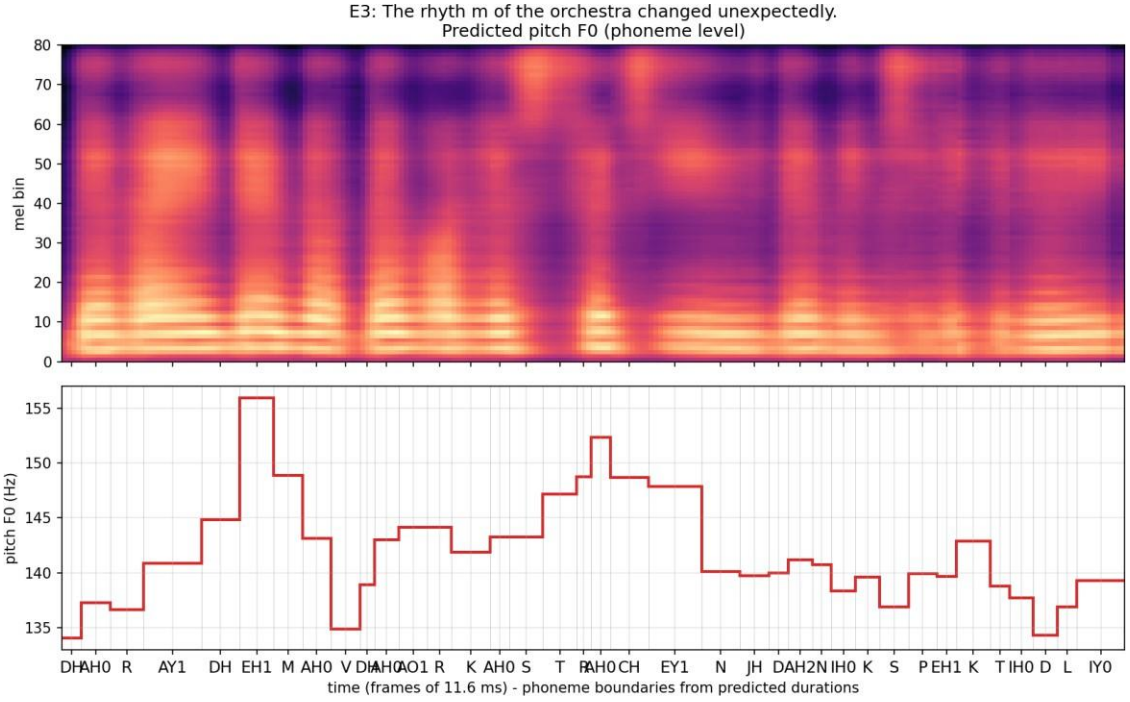


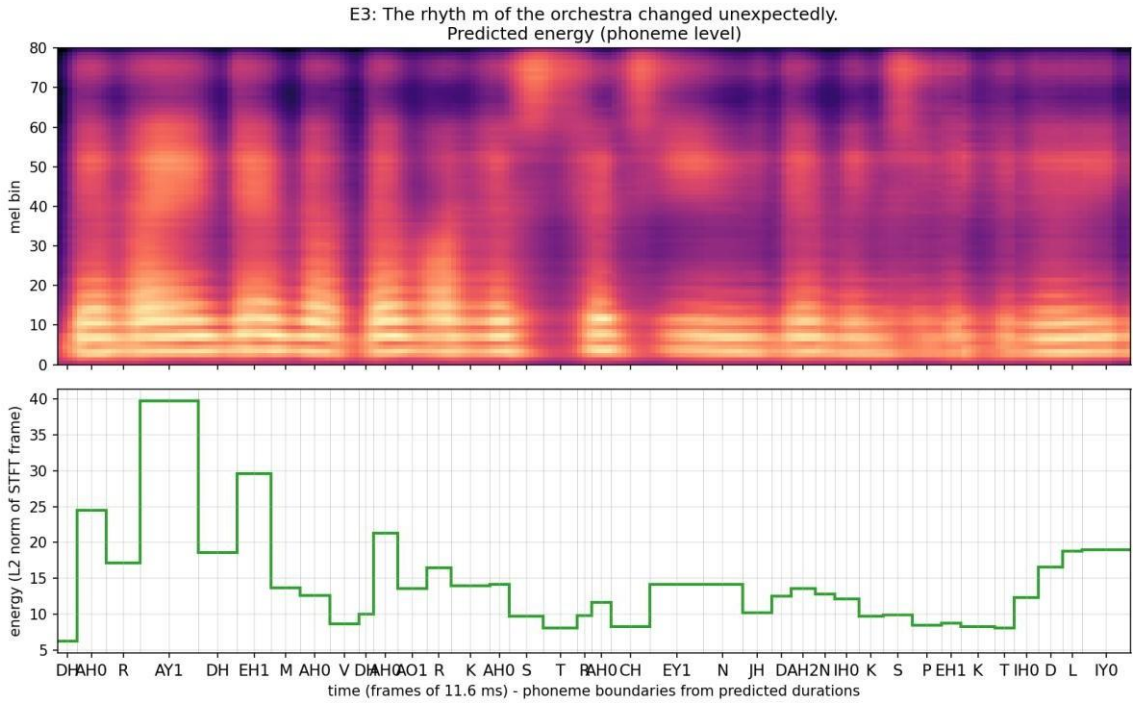


E3: The rhyth m of the orchestra changed unexpectedly.

Duration 2.57 s, 37 phones + 0 pauses; F0 134-156 Hz (mean 142); energy 6.26-39.70; longest: AY1 (139 ms), EY1 (128 ms), IY0 (116 ms)

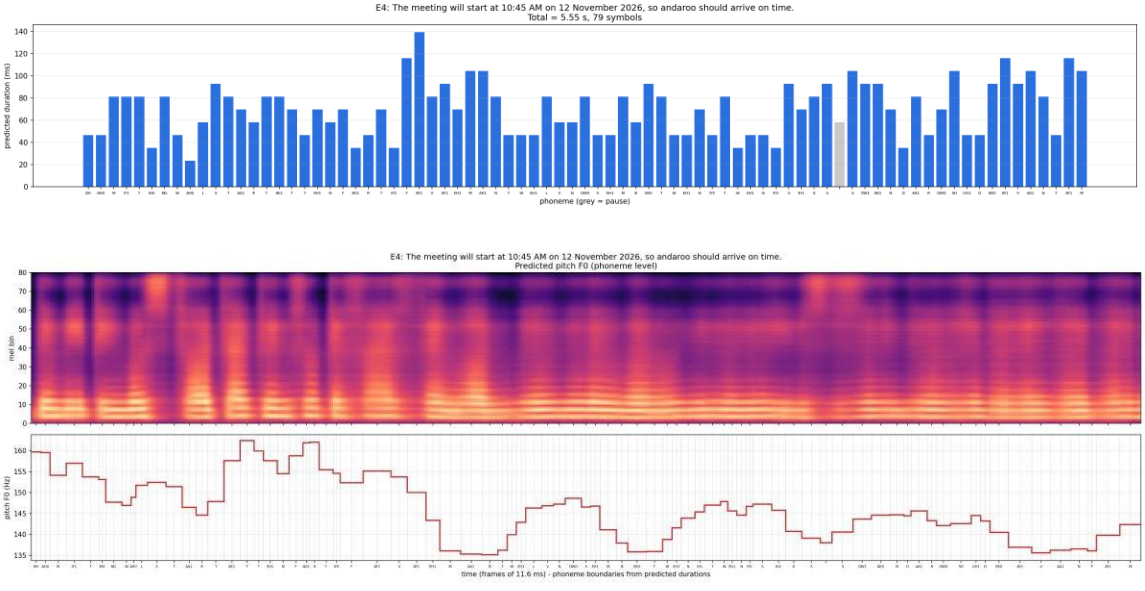

E3: The rhyth m of the orchestra changed unexpectedly.
Total = 2.57 s, 37 symbols
predicted duration (ms)
0
20
40
60
80
100
120
140
DH AH0 R AY1 DH EH1 M AH0 V DH AH0 AO1 R K AH0 S T R AH0 CH EY1 N JH D AH2 N IH0 K S P EH1 K T IH0 D L IY0
phoneme (grey = pause)

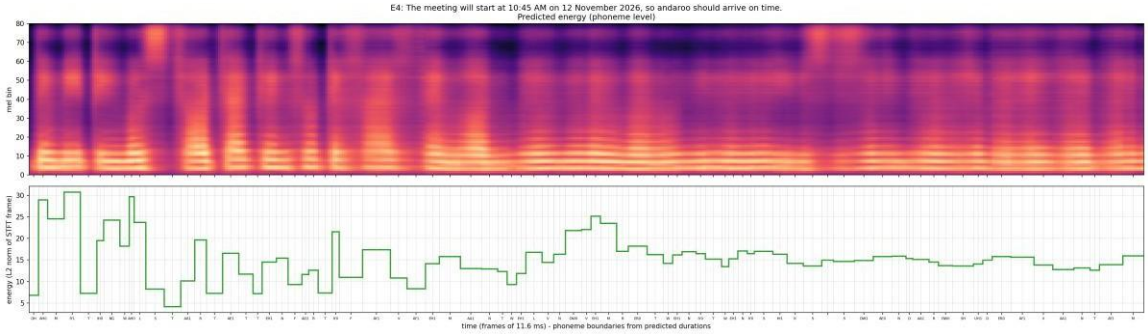

E3: The rhyth m of the orchestra changed unexpectedly.
Predicted pitch F0 (phoneme level)
mel bin
0
10
20
30
40
50
60
70
80
pitch F0 (Hz)
135
140
145
150
155
DH AH0 R AY1 DH EH1 M AH0 V DH AH0 AO1 R K AH0 S T R AH0 CH EY1 N JH DA AH2 N IH0 K S P EH1 K T IH0 D L IY0
time (frames of 11.6 ms) - phoneme boundaries from predicted durations



E4: The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Duration 5.55 s, 78 phones + 1 pauses; F0 135-162 Hz (mean 147); energy 4.17-30.73; longest: AY1 (139 ms), F (116 ms), AY1 (116 ms)




E4: The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.
Predicted energy (phoneme level)
mel bin
energy (L2 norm of STFT frame)
time (frames of 11.6 ms) - phoneme boundaries from predicted durations